Docusign Envelope ID: 79A712B3-7860-4CCD-B8B7-4DC38D886B9A

AVAREBIOTECH

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Field Case Study Report of **“Avare Biotech” app powered by MAKSA technology** on Frozen and Fresh Bull Semen (Sokoto, Nigeria, 2025)

Date: December 2025
Location: Sokoto, Nigeria

Conducted by: Abdulrauf Abdulrasheed (Sokoto, Nigeria)	**Verified by:** Avare Biotech Team (Dubai, UAE)
Abdulrauf is a veterinarian with six years of experience in clinical veterinary practice and livestock reproduction services (artificial insemination).	Maksim Mershiev, The CEO&Co-Founder of Avare Biotech
Signature: Signed by: 0D05A263EA14459...	Signature: Signed by: Maksim Mershiev 325996BF0B2B43A...

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1) EXECUTIVE SUMMARY / CONCLUSIONS

- **This case study clearly shows how essential it is to follow the video-recording instructions to obtain accurate results.** The microscope used allowed only 400x magnification, not the recommended settings, which changed how the sperm cells appeared and influenced the final measurements. Still, these videos were accepted as new data that help us improve the system.
- **MAKSA consistently detected key movement patterns across different breeds and sample qualities.** Even with non-standard video input, the app captured the overall dynamics of total and progressive motility, demonstrating that the technology works — and that video quality directly affects measurement accuracy.
- **The results highlight the value of standardized workflows for reliable semen evaluation.** These field tests gave Avare Biotech real-world evidence of where strict adherence to magnification, focus and sample preparation matters most. They also provided new material to further train and adapt MAKSA for additional magnifications, including 400x, supporting global scalability.

2) OBJECTIVE

To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech App for **frozen bovine semen** evaluation against conventional manual count methods.

MATERIALS AND METHODS

- Sample Type: Frozen–thawed bovine semen
- Dilution rate: 0
- Devices: Microscope + Smartphone (400x)
- Comparison Tools: Manual count
- Software: Avare Biotech App powered by MAKSA technology

Number of Videos: 5 per sample (1, 2, 3, 4, & 5 field of views).

SEMEN IDENTITY

- Breeds: Holstein Frisian, Simmental, and Brahma.

STRAW IDENTIFIERS

1. Holstein Frisian-BBIB SINGOSARI FH VF BOLTZ 322126 WW 0603.
2. Simmental-BBIB SINGOSARI SNI SIM NIGHTHAWK 617118 WW 0708.
3. Brahma-BBIB SINGOSARI SNI BRM SW UNIVERSE 42182 WW 0605.

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SPECIAL NOTE

- 200x magnification was lacking in the microscope used. Only the following objective lenses were available: 4x, 10x, 40x, and 100x. Thus, I used a 40x lens (400x magnification), which made the sperm cells bigger.

3) SEMEN ANALYSIS NOTE

Five different field views were taken from each sample**.** And here is the technical note of each sample:

- Holstein Frisian: rapid progressive motility was seen; the morphology appeared normal. In some field, the cells appear clumped together. The concentration of the sample was 50 million/ml; total motility and progressive motility were estimated at 75% and 30% respectively. The semen sample contains no red blood cells, leucocytes, or debris.
- Simmental: rapid progressive motility was also seen in this sample. The morphology appears normal. Cell clumping was seen in some field views. No debris or blood cells. The concentration of the sample is 55 million/ml; total motility and progressive motility were estimated at 70% and 28% respectively.
- Brahma: no rapid progressive motility with only a few cells seen moving sluggishly. In some field view, no motility at all. The concentration of the sample was 30 million/ml, and total motility and progressive motility were estimated at 10% and 1% respectively.

RESULTS TABLE

Bull's Name	**Key Semen Parameters**	**MAKSA**	**Manual Count**	**Difference, %**
HF1	Concentration M/ml	19.2	50	-61.60
	Total Motility %	35	75	-53.33
	Progressive Motility %	25	30	-16.67
HF2	Concentration M/ml	19.2	50	-61.60
	Total Motility %	35	75	-53.33
	Progressive Motility %	28	30	-6.67

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HF3	Concentration M/ml	24	50	-52.00
	Total Motility %	76	75	1.33
	Progressive Motility %	55	30	83.33
HF4	Concentration M/ml	14.4	50	-71.20
	Total Motility %	11	75	-85.33
	Progressive Motility %	9	30	-70.00
HF5	Concentration M/ml	20.8	50	-58.40
	Total Motility %	16	75	-78.67
	Progressive Motility %	12	30	-60.00
SIM1	Concentration M/ml	17.6	55	-68.00
	Total Motility %	40	70	-42.86
	Progressive Motility %	32	28	14.29
SIM2	Concentration M/ml	8	55	-85.45
	Total Motility %	44	70	-37.14
	Progressive Motility %	33	28	17.86
SIM3	Concentration M/ml	6.4	55	-88.36
	Total Motility %	38	70	-45.71
	Progressive Motility %	31	28	10.71
SIM4	Concentration M/ml	16	55	-70.91

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	Total Motility %	18	70	-74.29
	Progressive Motility %	13	28	-53.57
SIM5	Concentration M/ml	11.2	55	-79.64
	Total Motility %	41	70	-41.43
	Progressive Motility %	30	28	7.14
BRA1	Concentration M/ml	6.4	30	-78.67
	Total Motility %	11	10	10.00
	Progressive Motility %	8	1	700.00
BRA2	Concentration M/ml	8	30	-73.33
	Total Motility %	12	10	20.00
	Progressive Motility %	11	1	1000.00
BRA3	Concentration M/ml	1.6	30	-94.67
	Total Motility %	12	10	20.00
	Progressive Motility %	6	1	500.00
BRA4	Concentration M/ml	8	30	-73.33
	Total Motility %	12	10	20.00
	Progressive Motility %	11	1	1000.00
BRA5	Concentration M/ml	8	30	-73.33
	Total Motility %	15	10	50.00

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	Progressive Motility %	15	1	1400.00

*** Difference (%) = (MAKSA − Manual) ÷ Manual × 100**

4) DISCUSSION / EXPERT INSIGHTS

Sperm Concentration (M/ml)

General trend: MAKSA consistently reports much lower values than Manual counts. Patterns observed:

- Differences range from –52% to –95% across all bulls.
- SIM and BRA bulls show the largest negative discrepancies, often above –80%.
- HF bulls also show large negative differences, mostly around –60% to –70%.

Total Motility (%)

General trend: Highly variable; MAKSA often underestimates but occasionally overestimates. Patterns observed:

- HF bulls mostly show large negative differences (–53% to –85%).
- SIM bulls show moderate underestimation (–37% to –75%).
- BRA bulls show slight positive differences (+10% to +20%).

Progressive Motility (%)

General trend: Most variable parameter with extreme outcomes. Patterns observed:

- HF and SIM bulls show mild to moderate negative differences (–70% to +18%).
- BRA bulls show extremely high positive differences (+500% to +1000%) because the manual progressivity is very low (1%).

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5) CONCLUSIONS

- The results demonstrate how strongly video quality, magnification settings, and field preparation influence semen analysis outcomes. The use of 400x magnification — instead of the recommended workflow — changed the visual scale of the cells and led to measurable differences between manual estimates and MAKSA outputs. This confirms that standardized video protocols are essential for accurate comparison.
- Across HF and SIM samples, MAKSA consistently detected cell movement patterns, but concentration and motility values differed from manual readings due to the non-standard optical setup. These findings provide valuable new data that will be used to further adapt the algorithms for additional magnifications, including 400x.
- In Brahman samples, extremely low manual progressive motility (1%) made percentage-based differences appear disproportionately large. This highlights why proper video input and stable baselines are important for meaningful comparisons — and shows that MAKSA remains highly sensitive to actual motion captured in the video.

6) RECOMMENDATIONS

- Follow Avare's instruction on how to make proper videos for the app analysis to ger more accurate results.

APPENDIX

• Screenshot of the semen videos.
• MAKSA PDF report samples.

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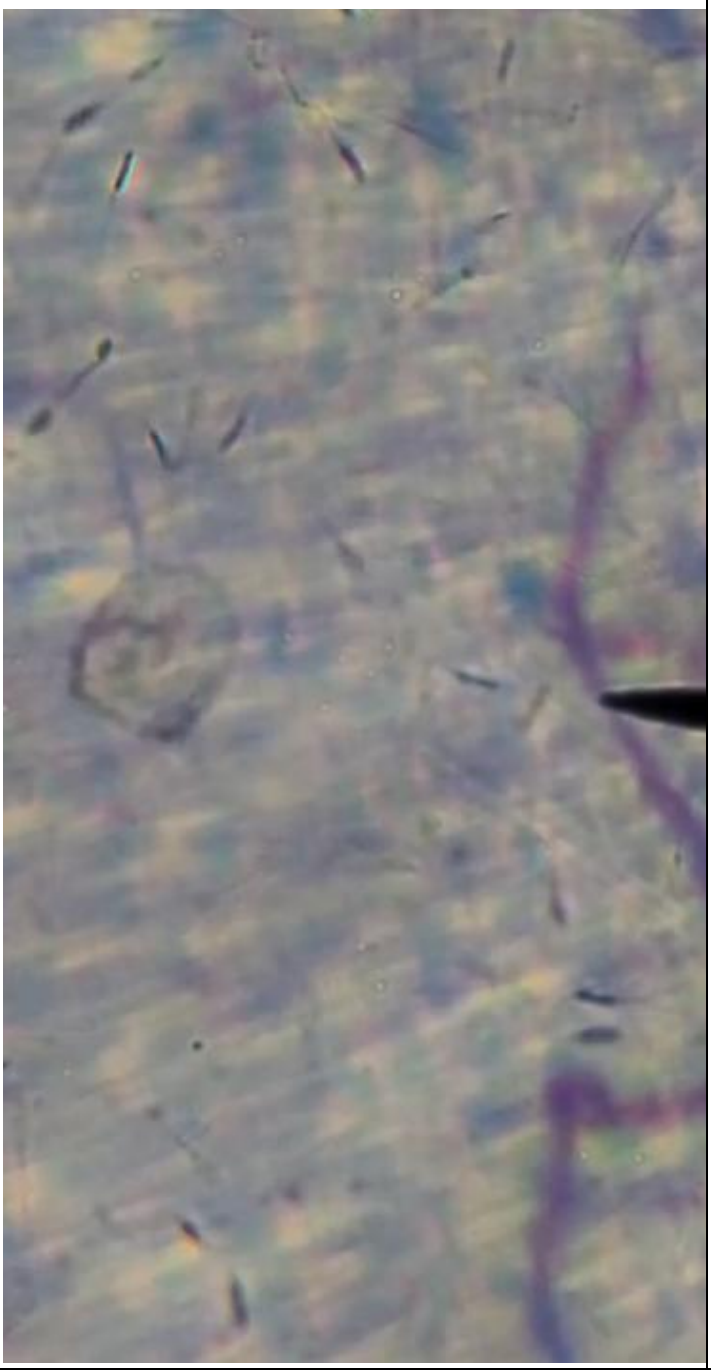
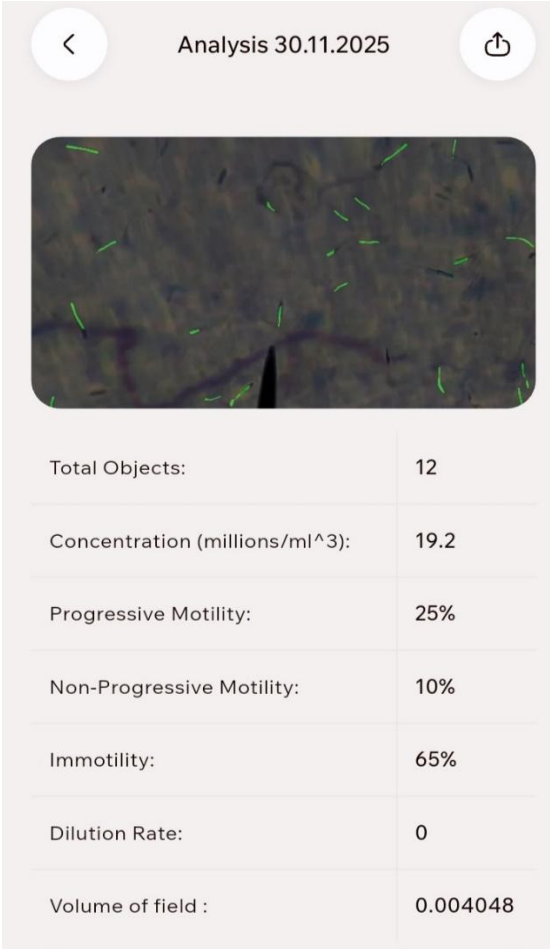
HF 1: Manual Count	HF 1: MAKSA values
	

Analysis 30.11.2025

Total Objects:	12
Concentration (millions/ml^3):	19.2
Progressive Motility:	25%
Non-Progressive Motility:	10%
Immotility:	65%
Dilution Rate:	0
Volume of field :	0.004048

HF 2: Manual Count	HF 2: MAKSA values

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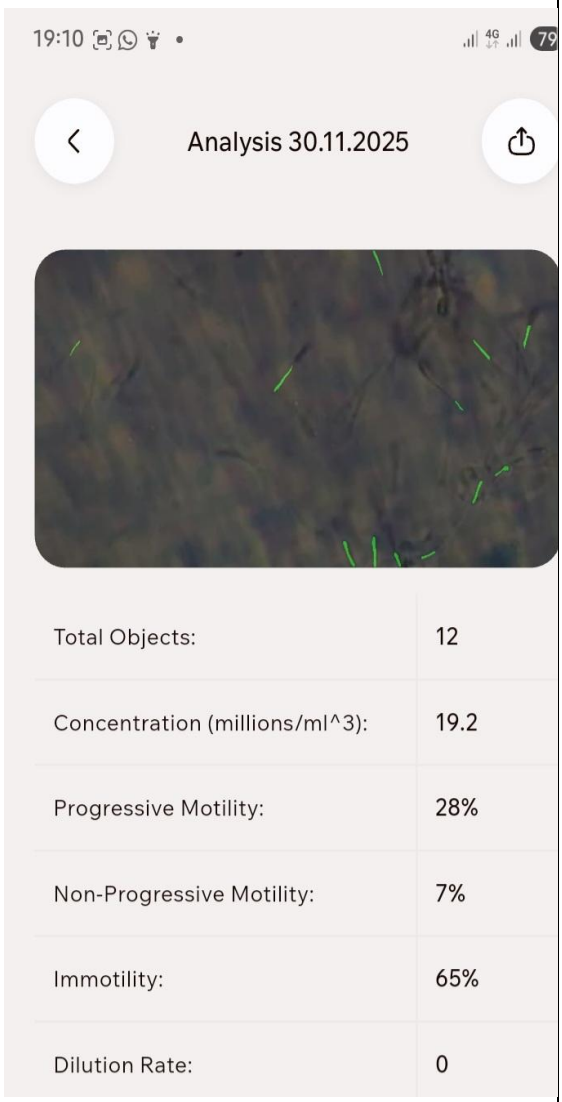

HF 3: Manual Count	HF 3: MAKSA values

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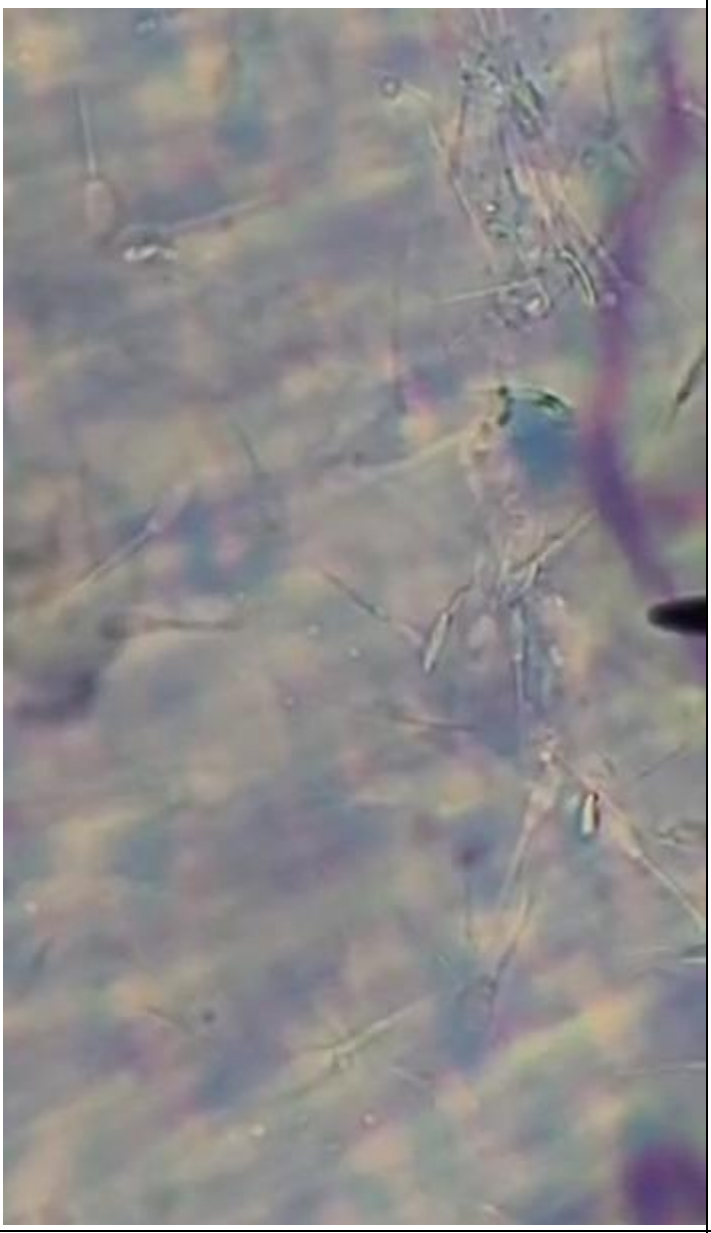

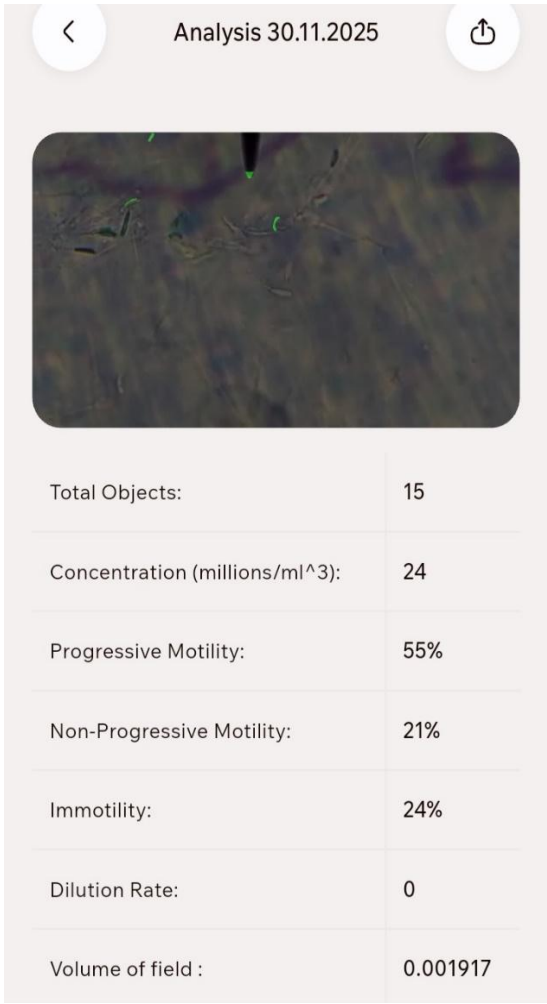

HF 4: Manual Count	HF 4: MAKSA values

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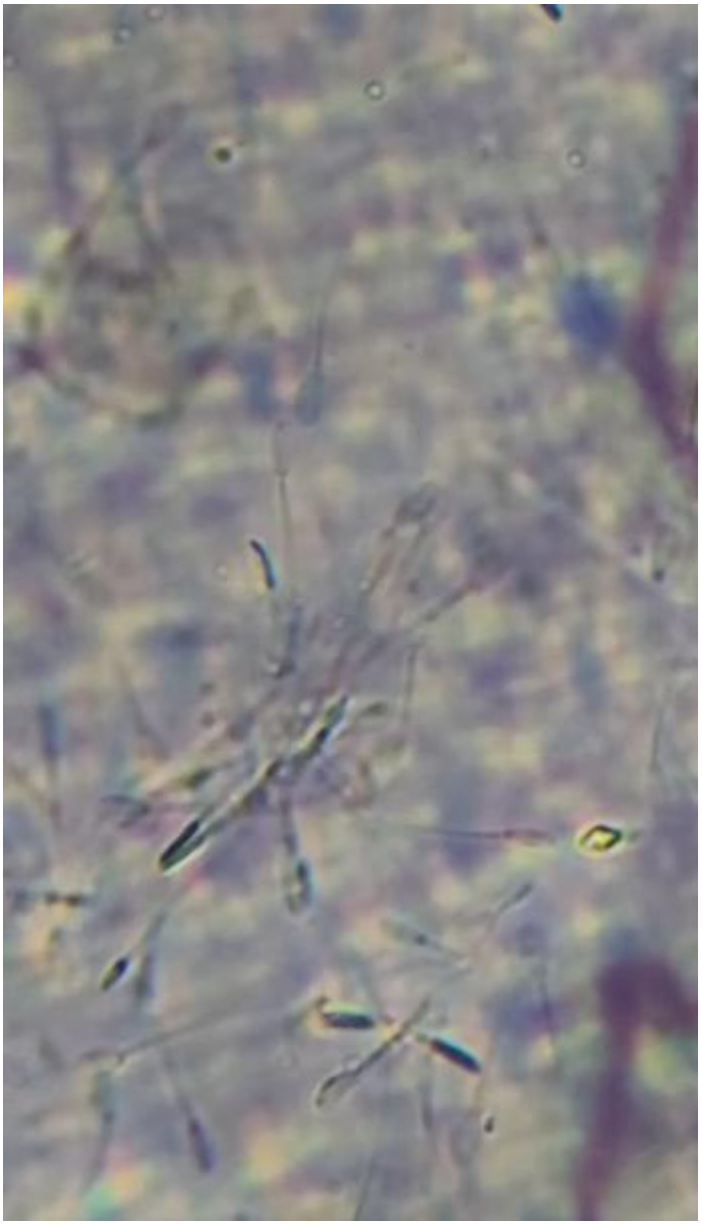

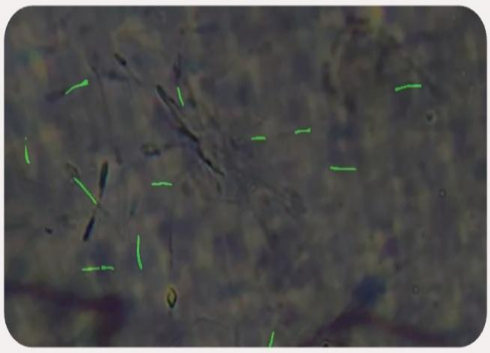

Total Objects:	9
Concentration (millions/ml^3):	14.4
Progressive Motility:	9%
Non-Progressive Motility:	2%
Immotility:	89%
Dilution Rate:	0
Volume of field :	0.004114

HF 5: Manual Count

HF 5: MAKSA values

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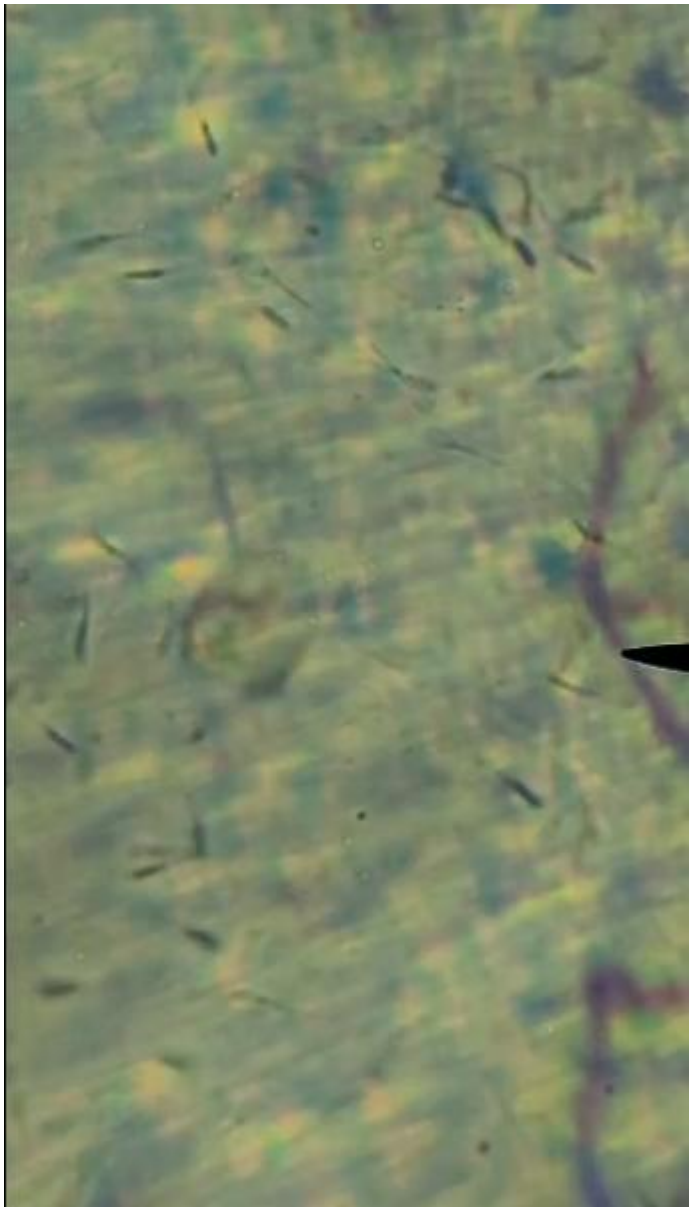
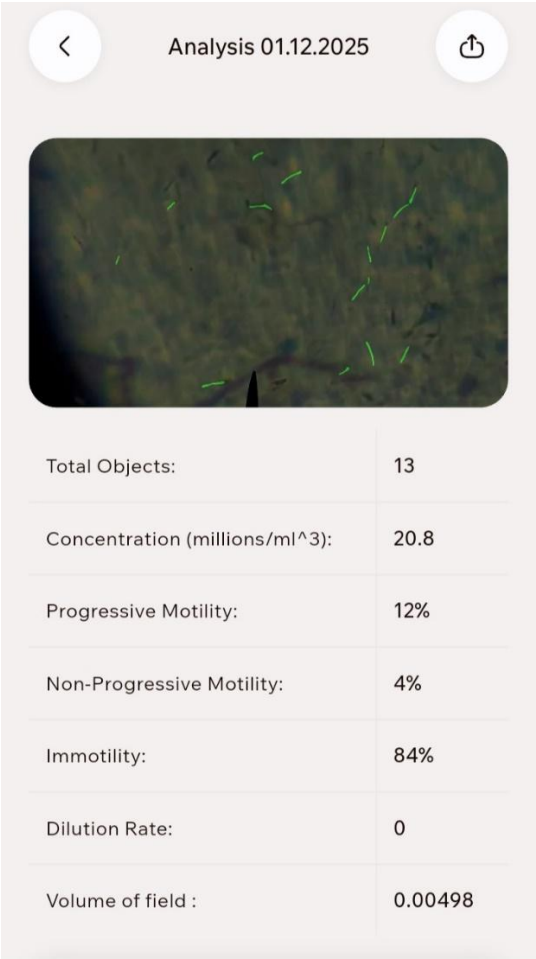
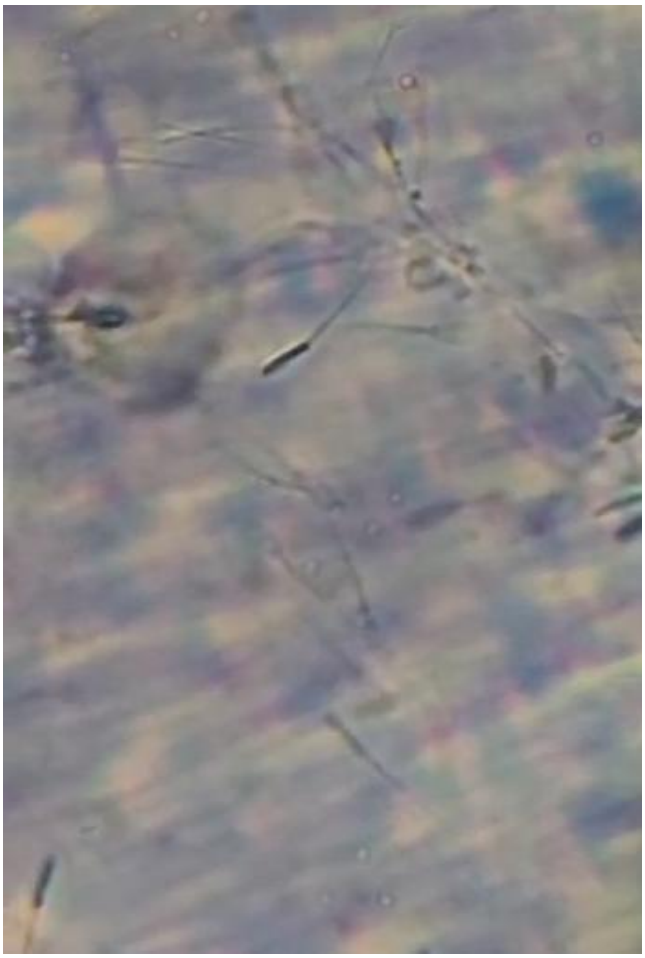
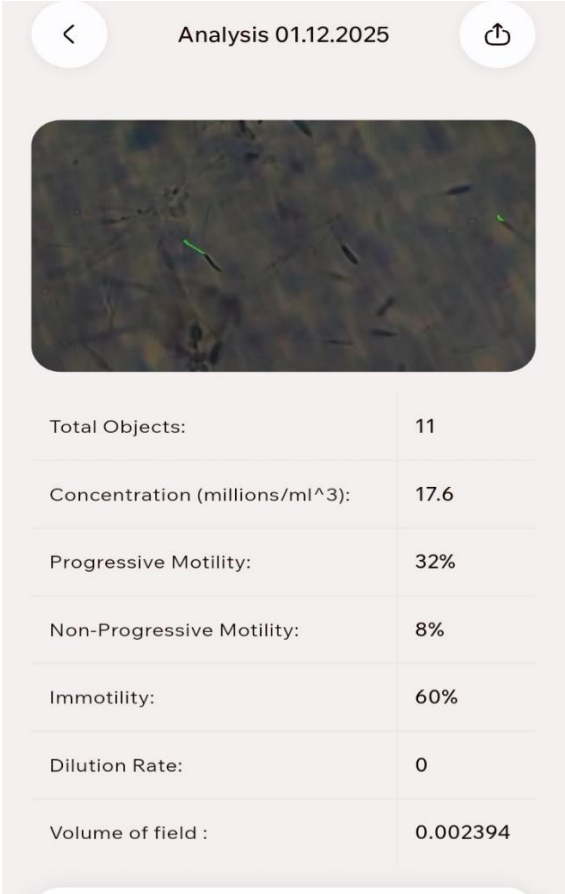




Analysis 01.12.2025

Total Objects:	13
Concentration (millions/ml^3):	20.8
Progressive Motility:	12%
Non-Progressive Motility:	4%
Immotility:	84%
Dilution Rate:	0
Volume of field :	0.00498

Docusign Envelope ID: 79A712B3-7860-4CCD-B8B7-4DC38D886B9A

SIM 1: Manual Count	SIM 1: MAKSA values
	
SIM 2: Manual Count	**SIM 2: MAKSA values**

Analysis 01.12.2025

Total Objects:	11
Concentration (millions/ml^3):	17.6
Progressive Motility:	32%
Non-Progressive Motility:	8%
Immotility:	60%
Dilution Rate:	0
Volume of field :	0.002394

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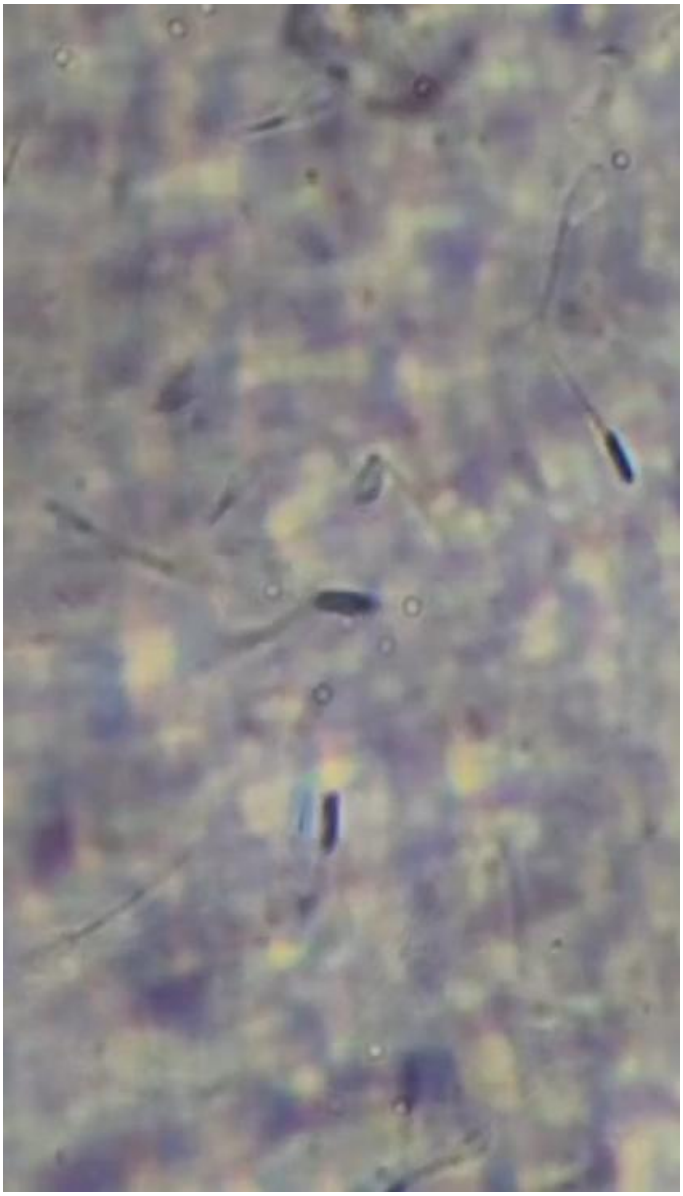

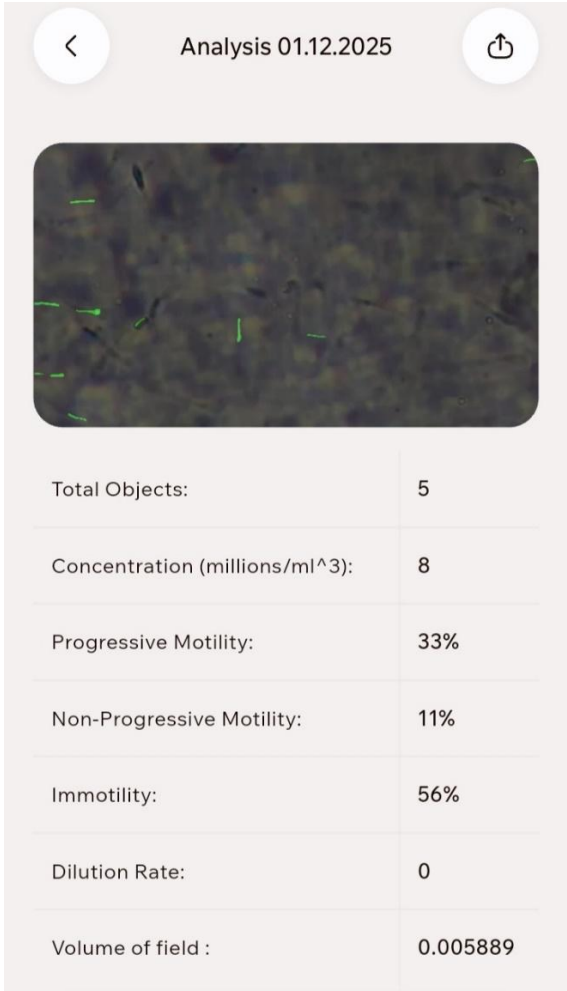

SIM 3: Manual Count	SIM 3: MAKSA values

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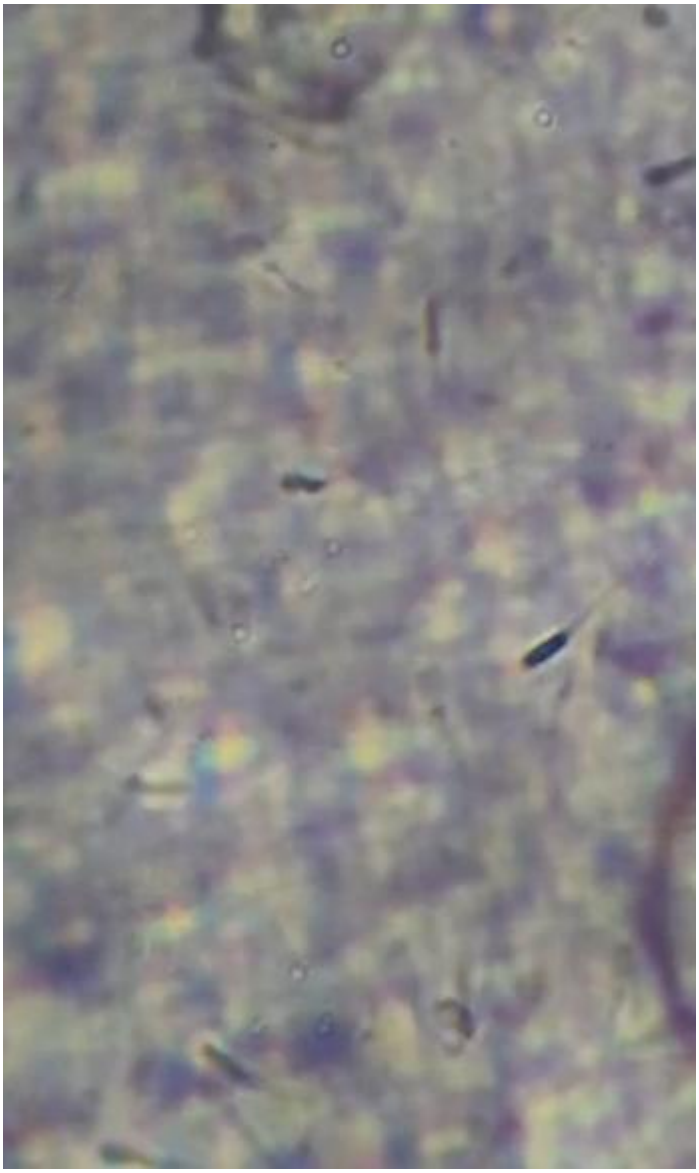

SIM 4: Manual Count	SIM 4: MAKSA values

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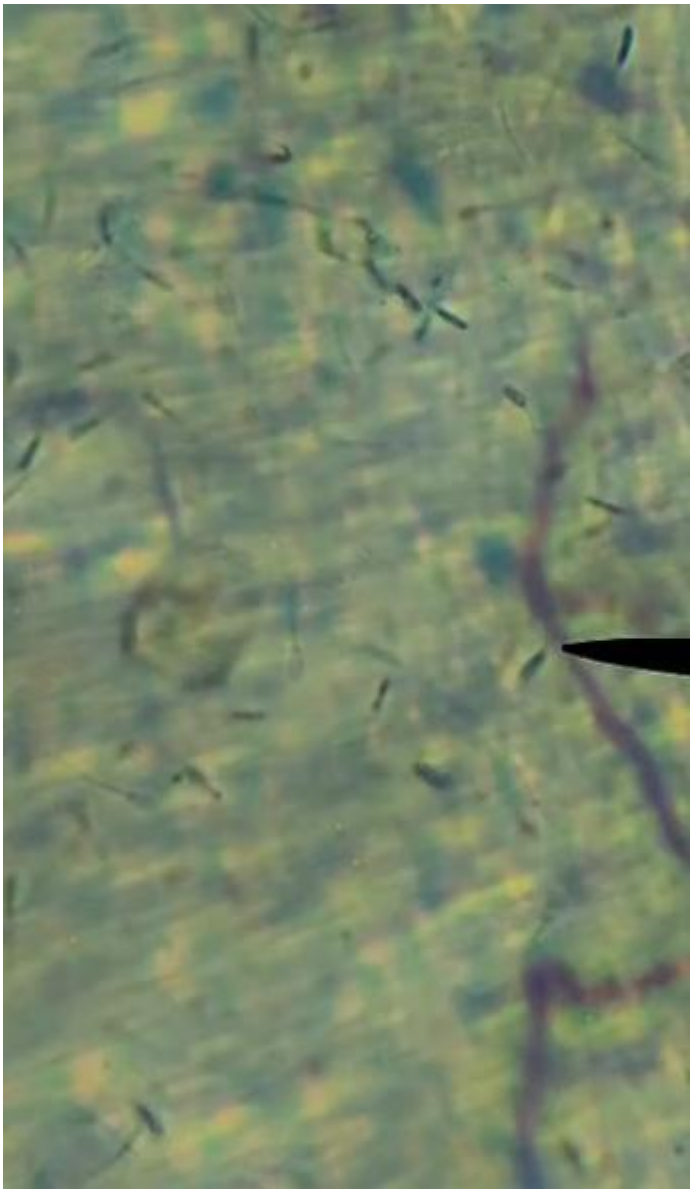

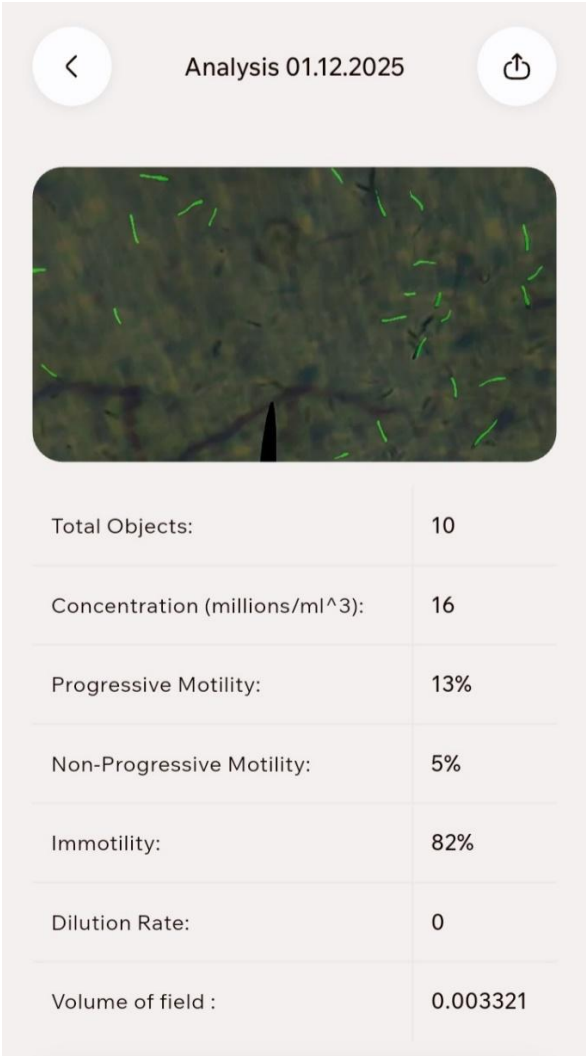

Analysis 01.12.2025

Total Objects:	10
Concentration (millions/ml^3):	16
Progressive Motility:	13%
Non-Progressive Motility:	5%
Immotility:	82%
Dilution Rate:	0
Volume of field :	0.003321

SIM 5: Manual Count

SIM 5: MAKSA values

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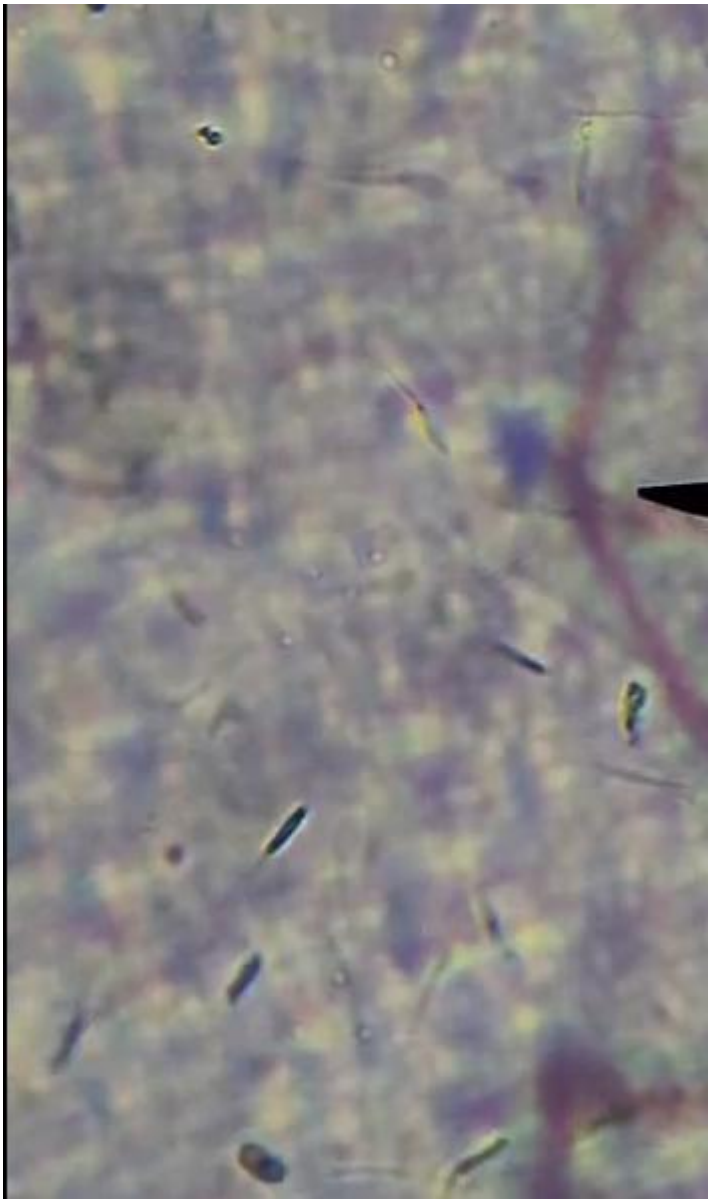
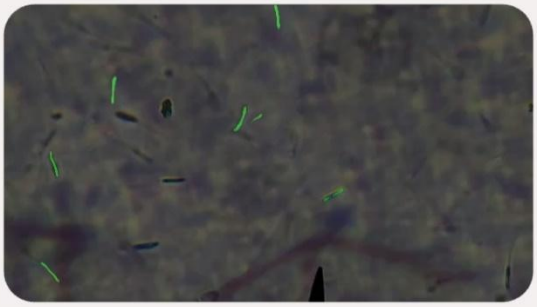


Analysis 01.12.2025



Total Objects:	7
Concentration (millions/ml^3):	11.2
Progressive Motility:	30%
Non-Progressive Motility:	11%
Immotility:	59%
Dilution Rate:	0
Volume of field :	0.003757

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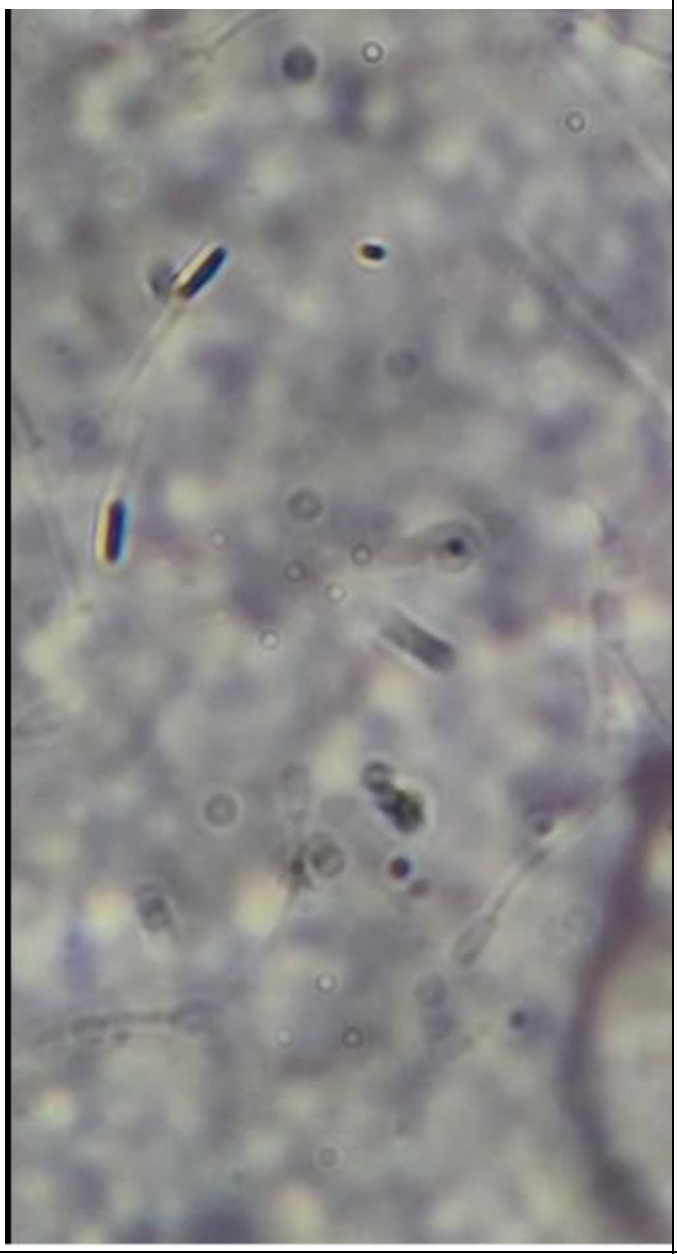
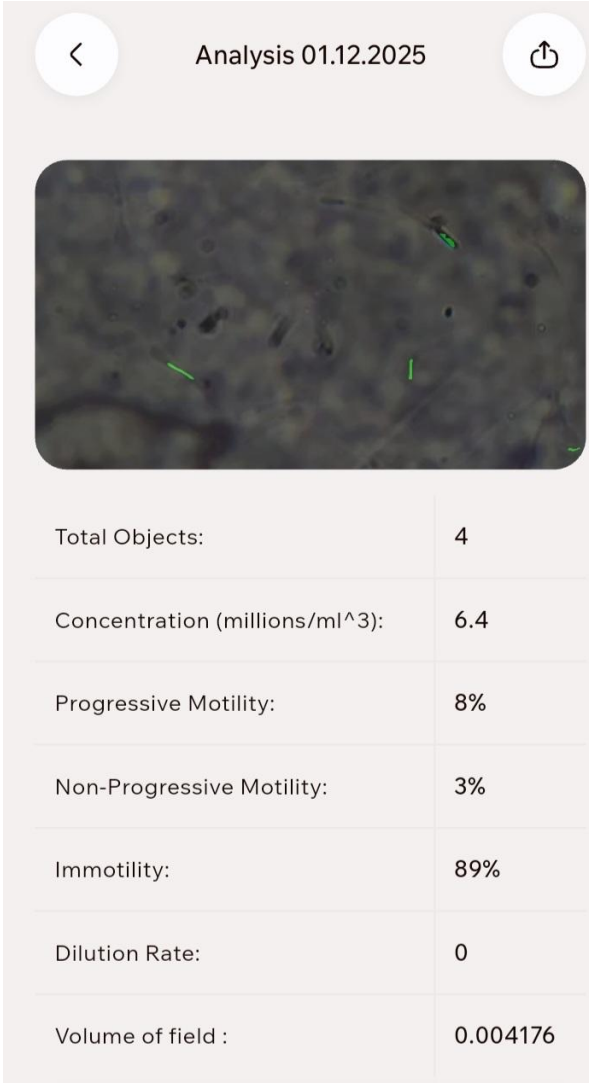
BRA 1: Manual Count	BRA 1: MAKSA values
	
BRA 2: Manual Count	**BRA 2: MAKSA values**

Analysis 01.12.2025

Total Objects:	4
Concentration (millions/ml^3):	6.4
Progressive Motility:	8%
Non-Progressive Motility:	3%
Immotility:	89%
Dilution Rate:	0
Volume of field :	0.004176

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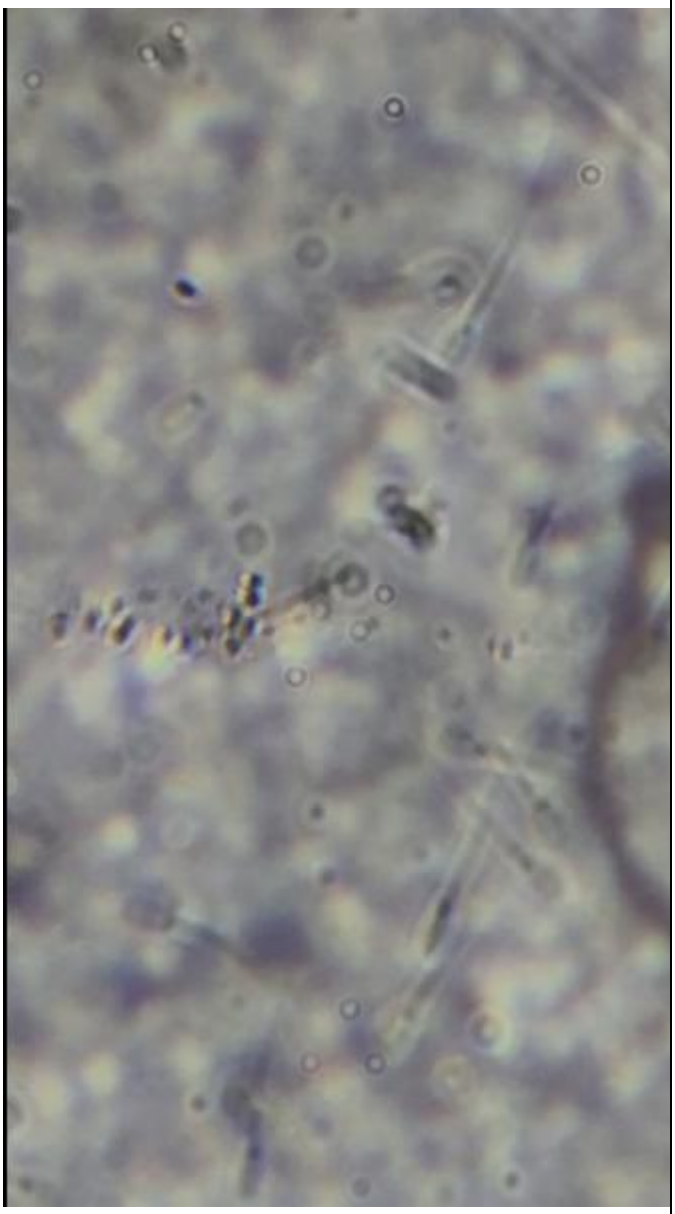
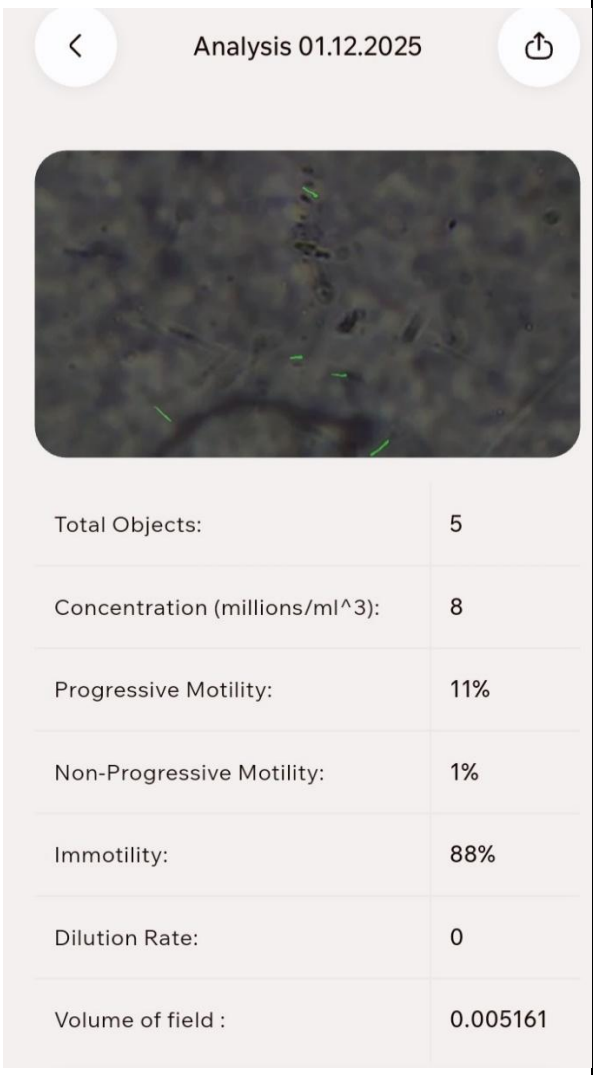


BRA 3: Manual Count



Analysis 01.12.2025

Total Objects:	5
Concentration (millions/ml^3):	8
Progressive Motility:	11%
Non-Progressive Motility:	1%
Immotility:	88%
Dilution Rate:	0
Volume of field :	0.005161

BRA 3: MAKSA values

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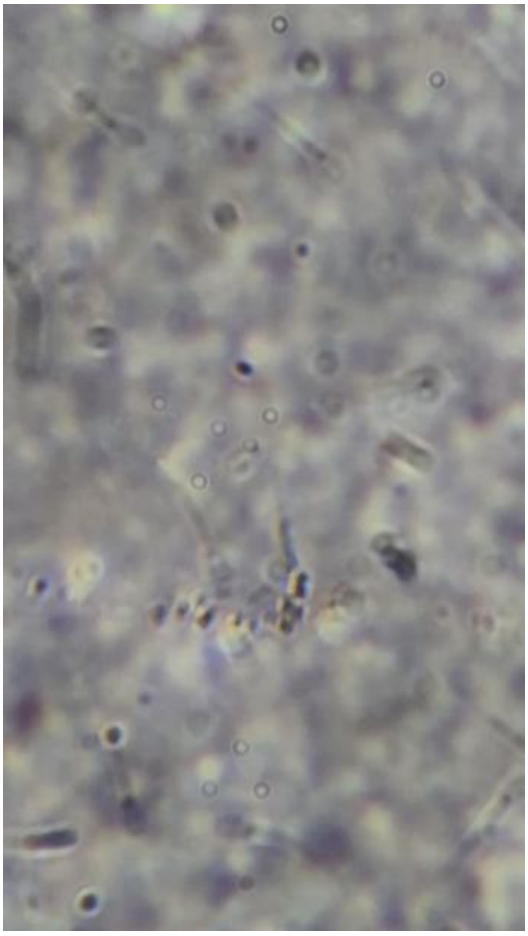


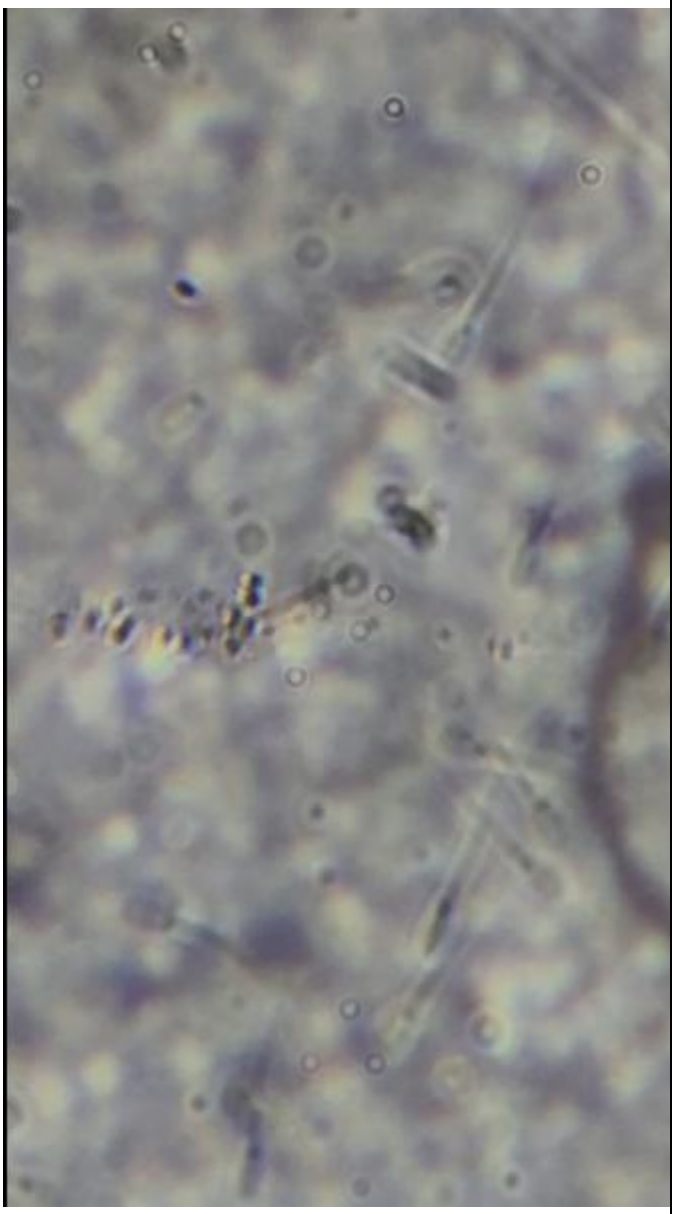
BRA 4: Manual Count

Analysis 01.12.2025

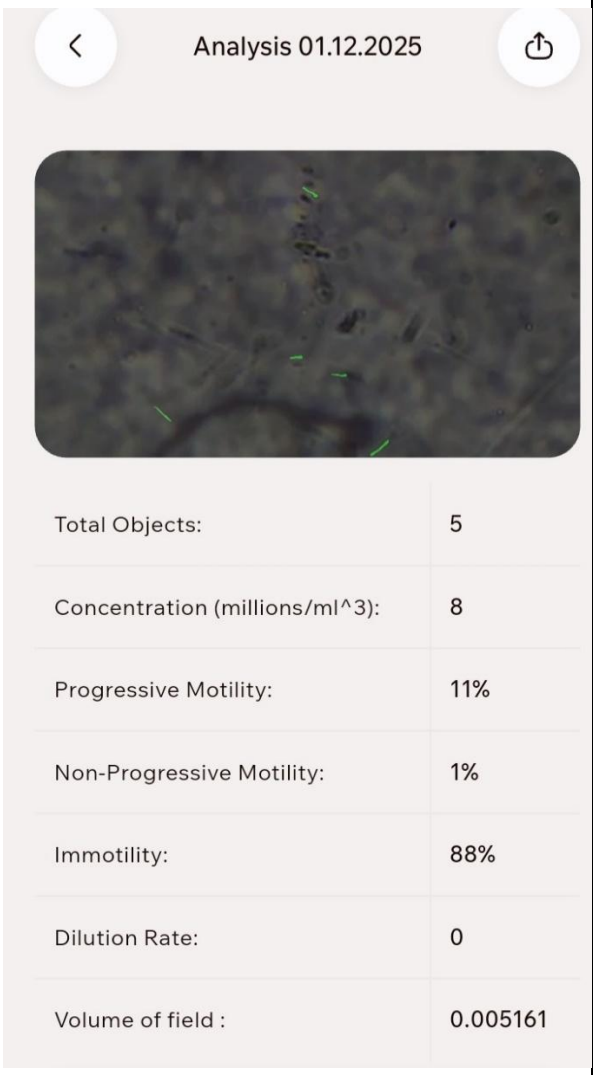
Total Objects:	1
Concentration (millions/ml^3):	1.6
Progressive Motility:	6%
Non-Progressive Motility:	6%
Immotility:	88%
Dilution Rate:	0
Volume of field :	0.004414

BRA 4: MAKSA values

Docusign Envelope ID: 79A712B3-7860-4CCD-B8B7-4DC38D886B9A



BRA 5: Manual Count



Analysis 01.12.2025

Total Objects:	5
Concentration (millions/ml^3):	8
Progressive Motility:	11%
Non-Progressive Motility:	1%
Immotility:	88%
Dilution Rate:	0
Volume of field :	0.005161

BRA 5: MAKSA values

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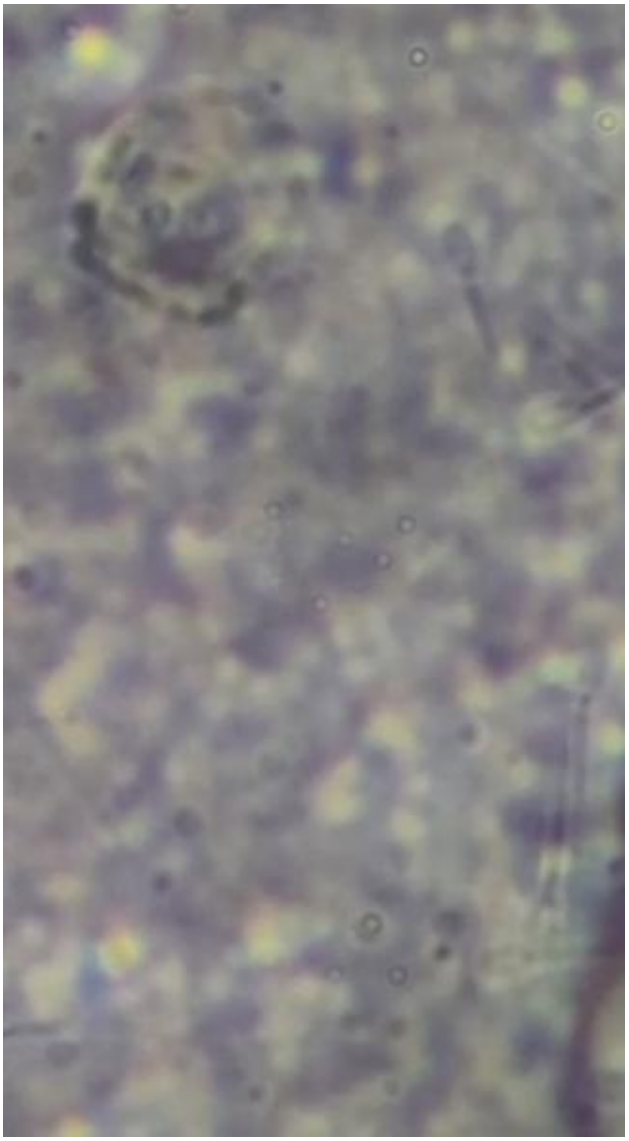
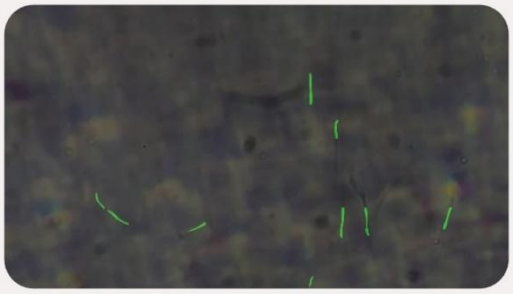


Analysis 02.12.2025



Total Objects:	5
Concentration (millions/ml^3):	8
Progressive Motility:	15%
Non-Progressive Motility:	0%
Immotility:	85%
Dilution Rate:	0
Volume of field :	0.00361